Setting Up your own VPN

Connecting two devices over the internet can be tricky due to firewalls, NAT, and other network configurations. A VPN (Virtual Private Network) allows you to create a secure connection between your devices over the internet, making it easier to access your Raspberry Pi from anywhere in the world (**or even on the Moon**). In this section, you will set up a VPN using TailScale, a popular and user-friendly VPN service based on the WireGuard protocol.

Creating a Tailscale Account and Installing Tailscale on your Laptop

1. On your laptop, go to Tailscale and sign up for a **free** account using your email.
2. Once you have created your account, log in to the Tailscale web interface.
3. Download and install the Tailscale client for your laptop's operating system (Windows, macOS, or Linux) from the Tailscale website (Download Page). Follow the installation instructions provided on the website.

Once you have installed Tailscale on your laptop, open the Tailscale application and log in using the account you just created. Your laptop should now be connected to the Tailscale VPN. You can verify this on your tailscale dashboard, where you should see your laptop listed as a connected device.

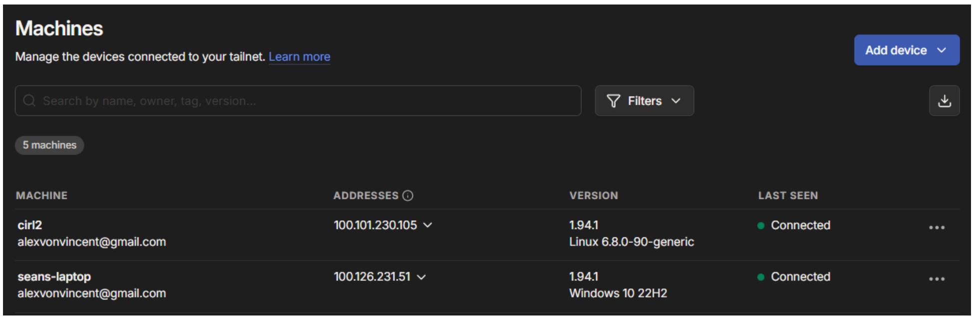


Figure 1: Tailscale Dashboard showing connected devices. In this case, I have quite a few devices connected! My laptop's name is seans-laptop.

> 📝 **Note: Getting your team connected to the Pi**
>
> Get your groupmates to also create Tailscale accounts and install the Tailscale client on their laptops. You will have to invite them to your Tailscale network later. This will allow everyone in your subgroup to access the Raspberry Pi remotely.
>
> You can have a maximum of 3 unique users per tailnet (i.e., tailscale network) on the free account. If you want more users, we suggest that your entire group create and share a Tailscale account for simplicity.

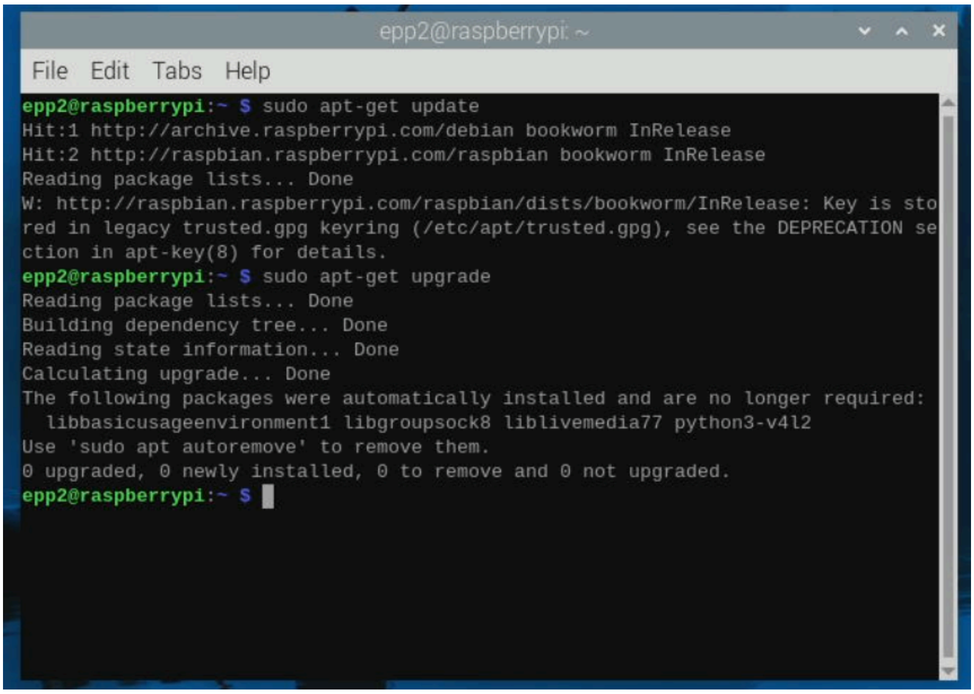
Setting up Tailscale on your Raspberry Pi

Now that you have Tailscale set up on your laptop, it's time to install and configure Tailscale on your Raspberry Pi. Make sure your Raspberry Pi is powered on and connected to the internet.

Open a terminal on your Raspberry Pi (you can use the terminal application in the Raspberry Pi OS desktop environment). Run the following commands:

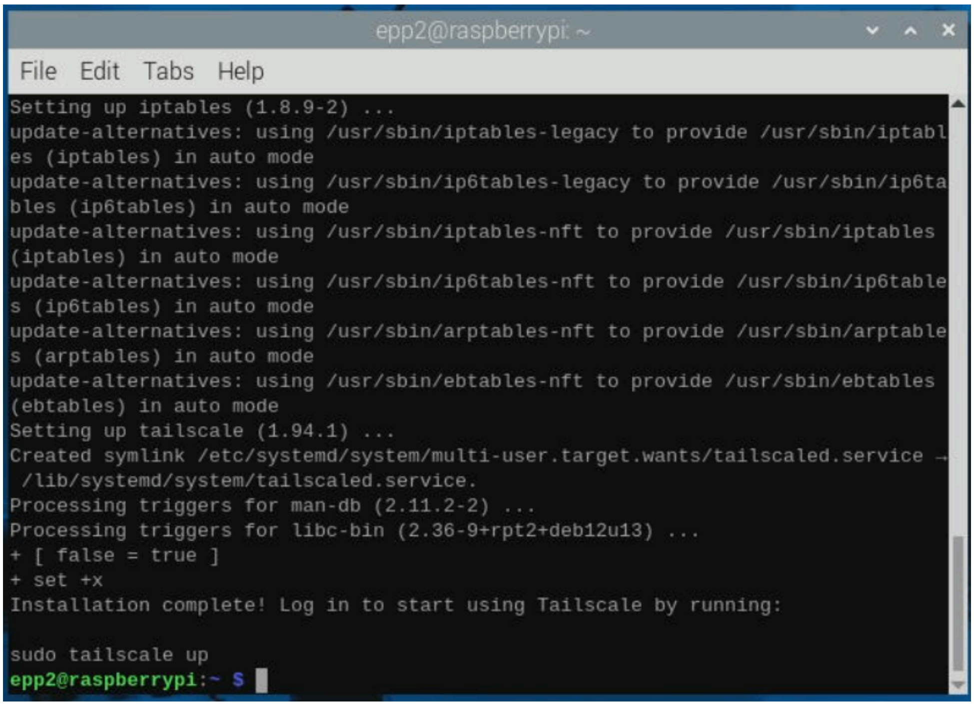
```
# Lines starting with '#' are comments, do not type them in
# These commands will update the package list and install any updates if available.
# Just in case the provided OS image is outdated
# or the previous update did not complete successfully.
```

```
sudo apt-get update
sudo apt-get upgrade
```



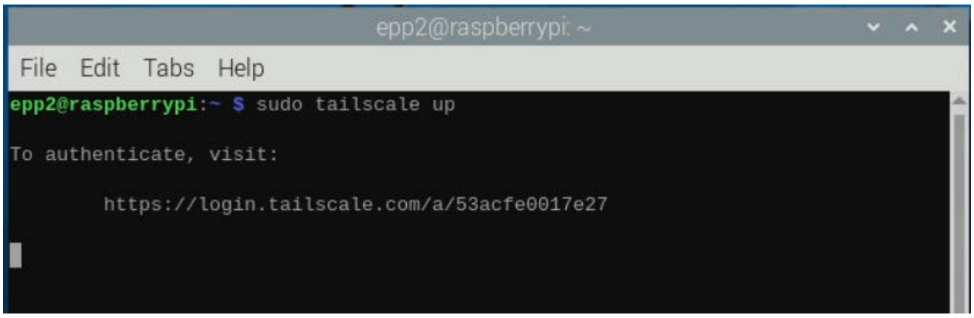
Next, install Tailscale by running the following command:

```
# We will download the official installation script from Tailscale
# (using the program curl) and execute it to install Tailscale on the Pi.
curl -fsSL https://tailscale.com/install.sh | sh
```



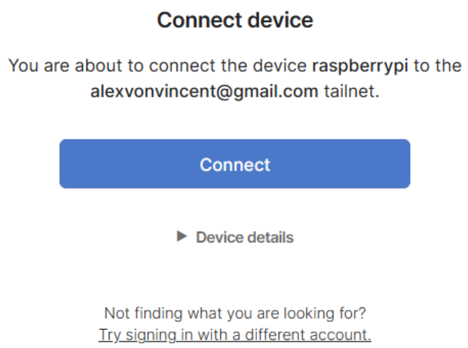
Once installed, you can attempt to start Tailscale and authenticate your Pi to your Tailscale account by running:

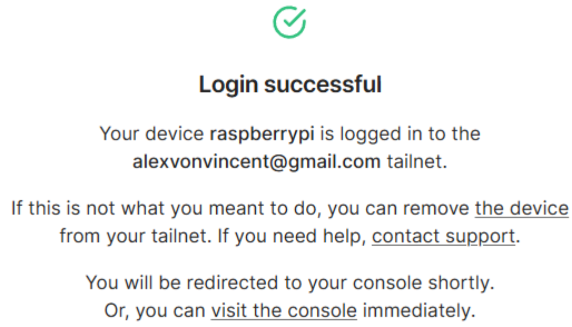
```
# This command will start the Tailscale service and provide you with a link to
# authenticate your Raspberry Pi to your Tailscale account.
# It should also set up to automatically start the Tailscale service on boot.
sudo tailscale up
```



Proceed to the link provided in the terminal output (Do not use mine! it will not work!) to authenticate your Raspberry Pi to your Tailscale account. You may need to log in to your Tailscale account if you are not already logged in.

Connect the Pi to your Tailscale network by clicking “Connect” on the web page.





If you followed the steps correctly, your Raspberry Pi should now be connected to your Tailscale VPN. You can verify this by checking the Tailscale dashboard on your laptop.

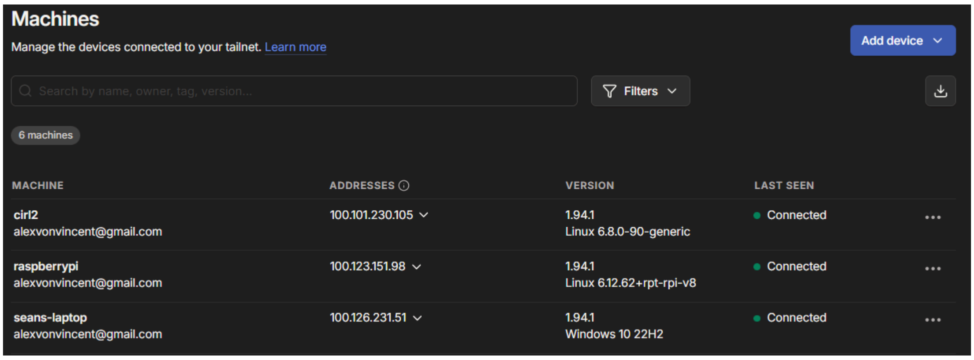
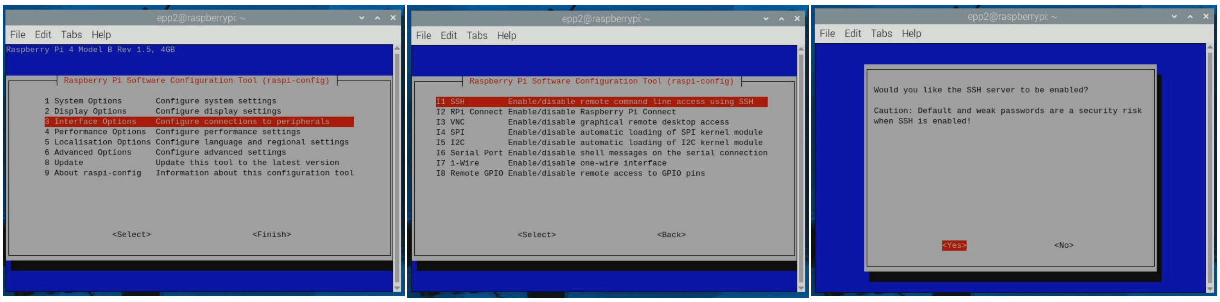


Figure 2: My Raspberry Pi is named raspberrypi.

Finally, to prepare for remote access, you need to enable the SSH server on your Raspberry Pi. Run the following command on your Raspberry Pi and navigate to “Interfacing Options” -> “SSH” -> “Enable”:

```
# This command allows access to many configuration options for the Raspberry Pi.
# We will use it to enable SSH.
sudo raspi-config
```



Testing the Tailscale Connection Via SSH (From your Laptop!)

> 📝 **Note**
>
> From this point, you theoretically don't need to keep the monitor, keyboard, and mouse connected to the Raspberry Pi anymore. For the true headless engineering experience, you can unplug them for the remainder of the lab! (Plugging them back in is also fine if you need to troubleshoot anything.)

Congratulations! You have successfully set up Tailscale on your Raspberry Pi and enabled the SSH server. Now, let's test the connection by SSHing into your Raspberry Pi from your laptop using the Tailscale IP address.

First obtain your Raspberry Pi's Tailscale IP address. You can find this on the Tailscale dashboard on your laptop.

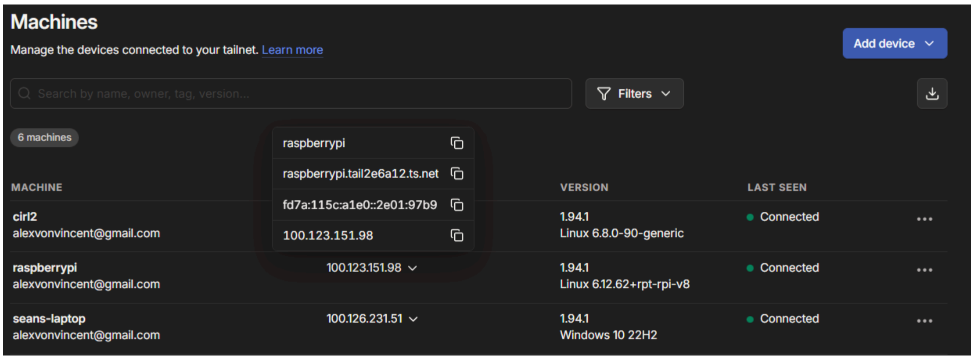


Figure 3: Finding the Tailscale IP address of the Raspberry Pi on the Tailscale dashboard. In this case, it is 100.123.151.98

Take note of the Tailscale IP address of your Raspberry Pi (it should be in the format 100.x.x.x). This address is unique to your Raspberry Pi within your Tailscale network. You will need this IP address to SSH into your Raspberry Pi from your laptop.

Open a terminal on your laptop and run the following command, replacing <TAILSCALE_IP> with your Raspberry Pi's Tailscale IP address and <USERNAME> with the username you set during the Raspberry Pi OS setup (default is "pi"):

```
# If you have taken CS1010 / CS1010E, you should be familiar with SSH.
# Just that now, we are using our very own VPN to connect to the Pi remotely.
ssh <USERNAME>@<TAILSCALE_IP>
```

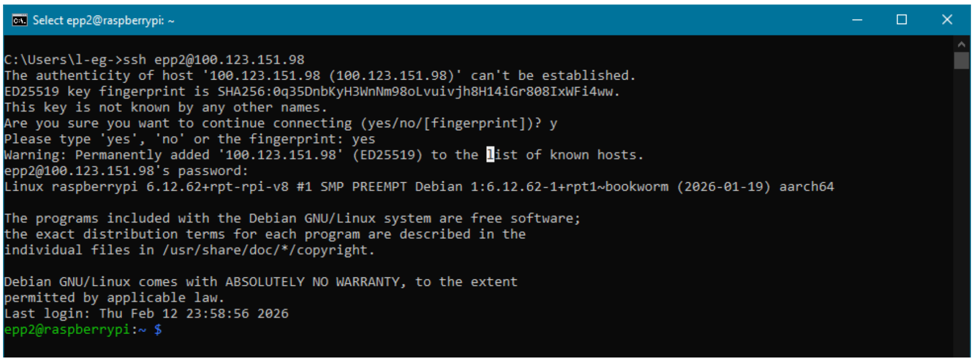


Figure 4: The username I set for my Pi is "epp2" and my tailscale IP is 100.123.151.98, so I ssh into `epp2@100.123.151.98`. I hope you remember your password!

If everything is set up correctly, you should now be logged into your Raspberry Pi via SSH! You can now run commands on your Raspberry Pi remotely from your laptop. You can now shut down your Pi to prepare for the next step.

```
# This command will shut down the Pi immediately.
sudo shutdown now
```

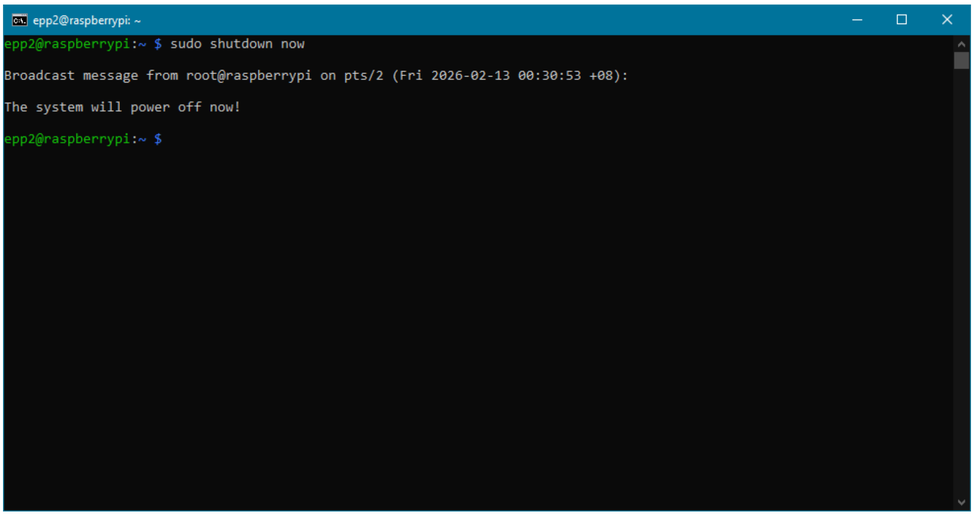


Figure 5: Make sure to shut down the Pi properly before unplugging power!

Setting up the Pi Camera

In this section, you will set up the Raspberry Pi Camera Module and take a photograph using it, then transfer the photograph to your laptop.

📝 **Note**

Security alert! After these steps, you've added a privacy sensitive sensor to your Raspberry Pi. Please cover your camera or unplug your Pi when not in use.

Connecting the Pi Camera Module

The Raspberry Pi Camera Module is a small camera that connects to the Raspberry Pi via a dedicated camera interface (CSI). It allows you to capture high-quality images and videos. We provide the Pi Camera 3, which is compatible with the Raspberry Pi 4. **Make sure that the Raspberry Pi is powered off before connecting the camera module to avoid any damage.**

Figure 6: Camera CSI port is the one closer to the 'center' of the board.

Figure 7: Metal contacts face towards the HDMI ports.

Figure 8: Fully seated camera cable with the clip securely holding it in place.

Follow these steps to connect the Pi Camera Module to your Raspberry Pi:

1. Locate the Camera Serial Interface (CSI) port on the Raspberry Pi. It is a small, flat connector located near the HDMI ports.
2. Gently lift the plastic clip on the CSI port to open it.
3. Insert the camera cable into the CSI port with the metal contacts facing towards the HDMI ports. Make sure the cable is fully seated in the port.
4. Push the plastic clip back down to secure the camera cable in place.

Testing the Pi Camera

Now that the Pi Camera Module is connected, it's time to test it out by taking a photograph. First, power on your Raspberry Pi by connecting it to a power source. Wait for it to boot up completely (about a minute).

Once the Pi is booted up, SSH into your Raspberry Pi from your laptop using the Tailscale IP address as described in the previous section.

The Raspberry Pi OS provides some pre-installed programs for you to interact with the camera. To test the camera connection, run the following command on your Raspberry Pi:

```
# This command will test if the camera is connected properly.
rpicam-jpeg --list-cameras
```

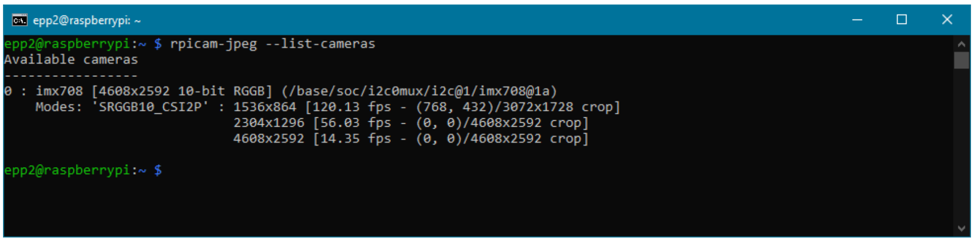
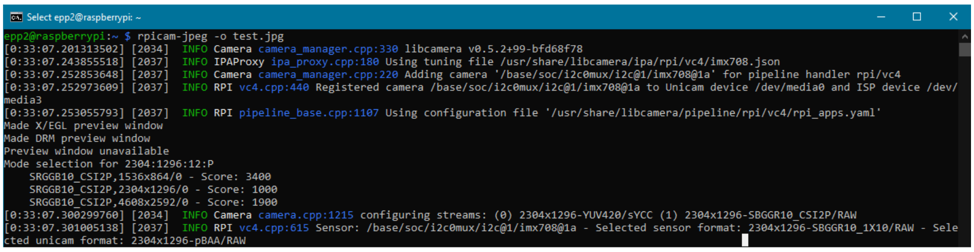


Figure 9: Output of `rpicam-jpeg --list-cameras`. If you see "0: imx708" listed, the camera is detected successfully!

Now that the camera is detected, you can take a photograph using the following command:

```
# This command will take a photograph and save it as 'image.jpg' in the current directory
# The rpicam-jpeg program has many camera options that you can explore later. You can check them using the '-h' flag
rpicam-jpeg --output test.jpg
```



Check that the photograph has been taken successfully by listing the files in the current directory.

```
# You should be familiar with this command from CS1010. It lists the files
# in the current directory. You should see 'test.jpg' in the output.
ls
```

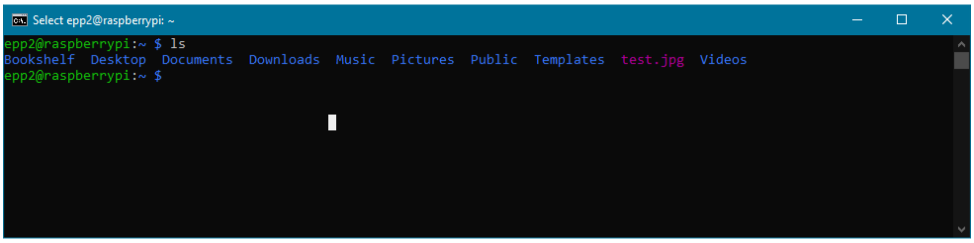
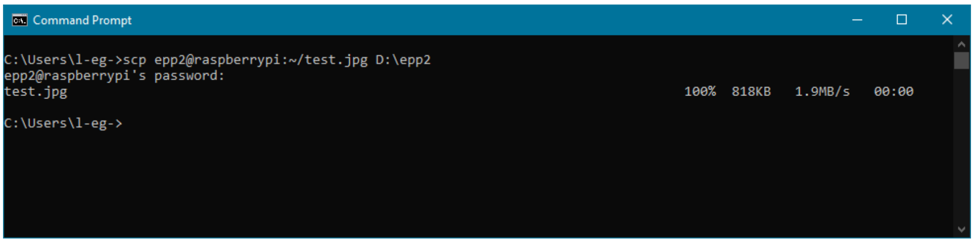


Figure 10: You can see 'test.jpg' in the output of `ls`, which means the photograph was taken and saved successfully!

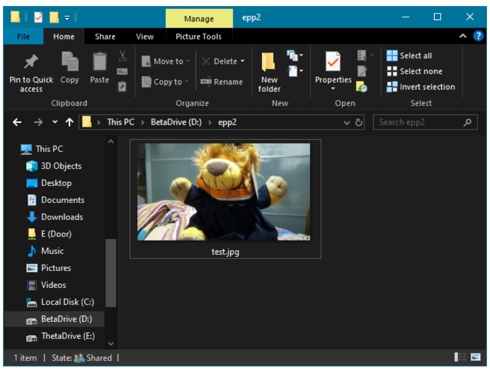
To really check that the photograph was taken successfully, you can transfer the photograph to your laptop using SCP (Secure Copy Protocol) over SSH.

Open another terminal on your laptop and run the following command, replacing <USERNAME> with your Raspberry Pi username, <TAILSCALE_IP> with your Raspberry Pi's Tailscale IP address, and <DESTINATION_PATH> with the path on your laptop where you want to save the photograph:

```
# This command will copy the 'test.jpg' file from your Raspberry Pi to your laptop.
Make sure to specify the correct path on your laptop where you want to save the file.
scp <USERNAME>@<TAILSCALE_IP>:~/test.jpg <DESTINATION_PATH>
```



Now if we open the photograph on your laptop, you should see the image captured by the Pi Camera Module!



Congratulations! You have successfully set up the Raspberry Pi Camera Module and taken a photograph using it.